

# NEW PRODUCTS

## TECHNOLOGY

### COMPONENTS

#### Processors

AMD has launched the EPYC 4005 series processors, delivering enterprise-grade performance in a scalable,



cost-efficient format for SMBs and IT providers. Built on the AM5 socket, they power rack, tower,

and blade servers with ease. The 16-core 4565P leads with 1.83x faster performance than Intel's Xeon 6300P.

AMD

<https://www.amd.com/en.html>

#### Microcontroller units

Renesas' RA8P1 MCU group redefines edge AI with breakthrough performance and intelligence. Built



to accelerate voice, vision, and real-time analytics, it enables

smarter automation and immersive interfaces. With dedicated AI acceleration and robust security, it is ideal for advanced HMI, surveillance, robotics, and conferencing, making complex AI tasks seamless at the edge.

Renesas Electronics Corporation

<https://www.renesas.com/en>

#### Digital controllers

Microchip expands its dsPIC33A DSC family with the dsPIC33AK512MPS512



and MC510, designed for high-speed, high-precision control in SiC/GaN power systems and

multi-motor applications. Hardware safety, crypto, and secure boot support safety-critical and secure embedded designs with faster development and fewer external components.

Microchip Technology

<https://www.microchip.com>

#### Central processing units

Intel's Xeon 6 series introduces three high-performance CPUs designed for



demanding AI and GPU-accelerated workloads. With faster memory, expanded PCIe lanes, AMX support,

and DGX B300 compatibility, Xeon 6 delivers scalable performance, reliability, and intelligent compute for modern AI infrastructures and data centres.

Intel

<https://www.intel.com>

#### Dual-MOSFET

ROHM's AW2K21 30V N-channel MOSFET redefines fast-charging power



design. With just 2.0mΩ ON-resistance in a 2mm<sup>2</sup> WLCSP package, it reduces board

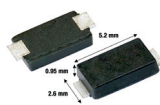
space by 81% and boosts efficiency by 33% less resistance than standard MOSFETs. Suitable for smartphones, wearables, and compact devices, its dual-MOSFET common-source design supports bidirectional protection.

ROHM Semiconductor

<https://www.rohm.com>

#### SiC diodes

Vishay's new Gen 3 SiC Schottky



diodes boost power efficiency and electrical isolation in a sleek, ultra-slim form

factor. Ideal for compact, high-voltage systems, they enable fast switching

and thermal stability, which are key for advanced energy, medical, and industrial applications where space, speed, and insulation are most critical.

Vishay Intertechnology, Inc.

<https://www.vishay.com/>

#### Smart sensor

Motorola HALO Smart sensor, launched in India by Arya Omnitalk, offers



real-time safety insights without compromising privacy. Ideal for schools, hospitals, and public infrastructure, it replaces

traditional cameras with 16 onboard sensors to detect vaping, air quality, aggression, gunshots, and more.

Arya Omnitalk

<https://aryaomnitalk.com>

#### TPMS sensor

Bosch's SMP290 is the first fully integrated BLE-enabled TPMS sensor, combining pressure, temperature, and



acceleration sensors, a microcontroller, and Bluetooth in a single MEMS unit. It simplifies

installation, cuts wiring, and lasts up to 10 years. Manufacturers save space and cost by integrating it with Bluetooth systems, such as keyless entry, enhancing safety, fuel efficiency, and regulatory compliance across all vehicle types.

Bosch

<https://www.bosch.in>

#### Entry-level microcontroller

GigaDevice has launched the GD32C231 series, high-performance entry-level



MCUs built on the ARM Cortex-M23 core. Ideal for smart appliances,

BMS, and handhelds, it offers up to 64kB flash memory with ECC, advanced